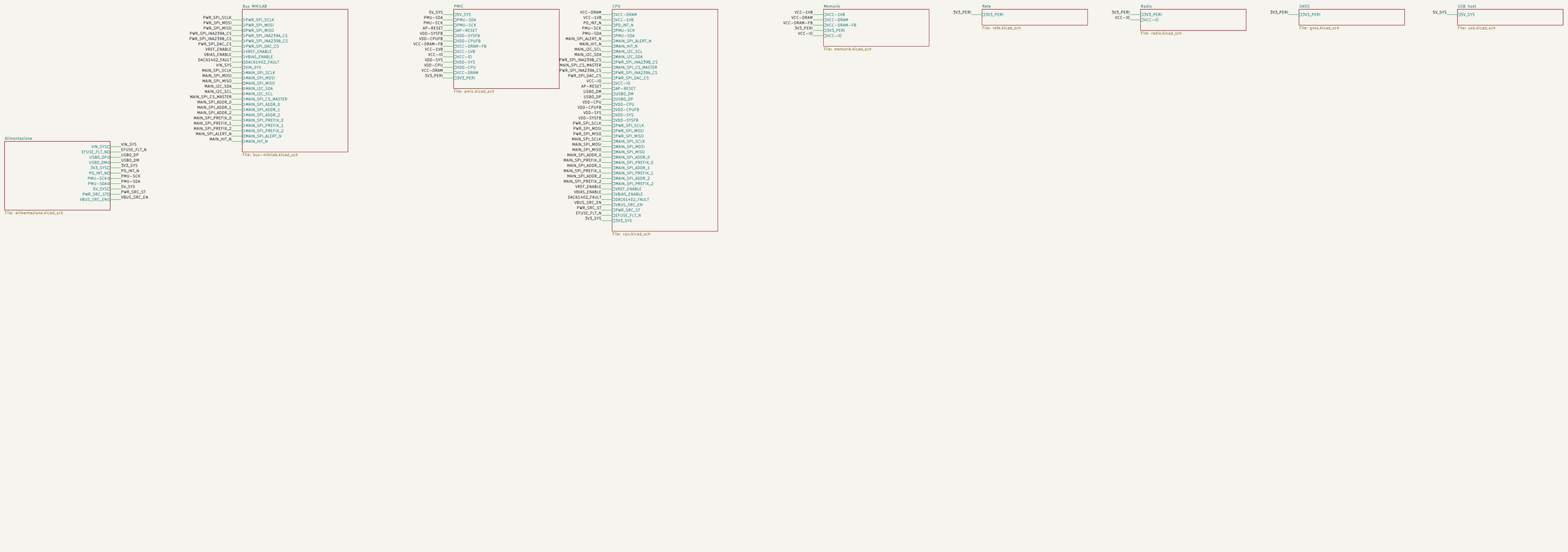


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KiCad E.D.A. 10.0.4 | Id: 1/10

CPU: Allwinner H616 (TFBGA-284, 0.65 mm). Ball map solo dal datasheet H616 V1.0 (tabella 4-2, verificata sulla figura 7-1). Serie 3xx.

AC_VRA1: la tabella dice D17, la figura 7-1 e il footprint EasyEDA dicono C17 -> usato C17. Mappa completa dei pin in docs/pinmap-h616.md.

Net DRAM_*/EMMC_*/SD_*/WL_*/BT_*/RGMII_*/PHY_*/GNSS_*/WIFI_*/USB1_*/SPI1_* globali verso gli altri fogli.

Clock: quarzo 24 MHz su DXIN/DXOUT (18 pF, 0R in serie su DXOUT). DXLDO_OUT 220 nF. L'H616 non ha pin per il quarzo 32 kHz: X32KFOUT (PG10) e' derivato internamente.

RESET (dominio VCC_PLL 1.8 V) dal PWROK dell'AXP313A + tasto: FEL a GND con tasto (pull-up 10k a VCC-IO). JTAG_SEL, PLLTEST, LRADC, REXT liberi.

J301: 1 = TX del SoC (PH0), 2 = RX del SoC (PH1), 3 = GND. Livelli 3.3 V.

Disaccoppiamento vicino alle alimentazioni (da rifinire sui reference Orange Pi Zero2/Zero3 e in layout).

Analogici (codec e AC300 non usati): VRA1/VRA2 470 nF verso AGND, AGND a GND con 0R, AC_VRA1 1 uF, ZQ del controller DRAM 240R 1% a GND.

SPI1 (PH6 CLK, PH7 MOSI, PH8 MISO) condiviso tra BUS_POWER e MAIN, chip select con cs-gpio. 33R serie per ramo. CLK/MOSI/MISO: EEPROM MAC locale su SPI1_* (foglio Rete).

Le MISO sono due rami unisconi dopo i 33R: tutti gli slave (INA239, DAC61402, bus MAIN) devono tenere SDO in alta impedenza quando CS alto. ADDR/PREFIX: 100R serie.

Livelli di riposo: CS_MASTER 3V3 (10k), VREF_ENABLE basso (10k a GND), VBIAS_ENABLE basso (100k a GND), ALERT# e HIT# 4.7k a I2C del bus 2.2k. FAULT 10k.

ESD bassa capacita TPD4E05U06DQA (0.5 pF, 5.5 V) sulle linee logiche di J1/J2, vicino ai connettori. Nessuna protezione su V_BIAS e riferimenti.

S_TWI0 (PL0/PL1, 1.8 V): AXP313A 0x36, FUSB302B 0x22, TCA6416A 0x20. Pull-up 2.2k a VCC-1V8. Gli open-drain su PC (1.8 V) hanno pull-up a VCC-1V8.

TCA6416A: VCC1 = VCC-1V8 (lato I2C), VCCP = VCC-IO (porte a 3.3 V). Ai reset tutte le porte sono ingressi: i pull-up/pull-down esterni fissano lo stato sicuro.

Porte: P00 VREF_ENABLE, P01 VBIAS_ENABLE, P02 DAC61402_FAULT (in), P03 USB1_VBUS_EN, P04 VBUS_SRC_EN, P05 PWR_SRC_ST (in), P06 PHY_RST_N, P07 PHY_INT_N (in), P10 WL_REG_ON, P11 BT_REG_ON, P12 BT_WAKE, P13 BT_HOST_WAKE (in), P14 WL_HOST_WAKE (in), P15 SD_DET (in), P16 PHY_EE_CS_N, P17 EFUSE_FLT_N (in).

R336: 3V3_SYS (VDD del FUSB302B) da VCC-IO.

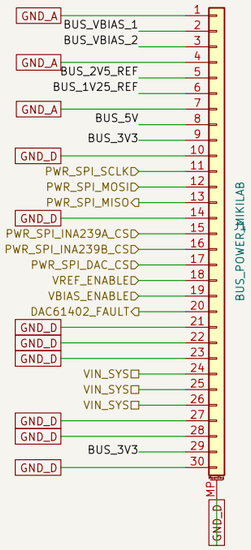
J1 BUS_POWER 30 pin (Hirose FH12-30S): pin 1-22 come J3 del modulo Power, 24-26 VIN_SYS 12 V, 29 secondo 3V3. Rail BUS_* passanti verso J2, non collegati ai rail della scheda.

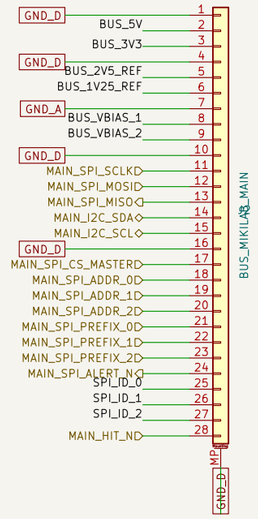
VIN_SYS -> eFuse TPS259470 -> buck TPS56637 12->5 V (foglio Alimentazione). BUS_5V va solo a J2 pin 2.

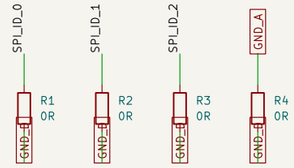
J2 MAIN 28 pin secondo BPS-MIKILAB-2026-REV5.1 (pin 28 = HIT#) (spec: XUNPU FPC-05F-28PH20; footprint H616:FPC-05F-28PH20 da EasyEDA C2856807).

Sul foglio CPU: SPI1 condiviso BUS_POWER/MAIN con 33R serie per ramo, 100R su ADDR/PREFIX, pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C, 10k sui CS, ESD TPD4E05U06.

R4 0R: ponte GND_A-GND_D, DNP di default (sul modulo Power GND_A e GND_D sono gia la stessa net). R1-R3: ID iniziale 000 (scheda = J_OUT, testa della catena).







Sheet: /Bus MIKILAB/
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KiCad E.D.A. 10.0.4		Id: 3/10

1) VIN_SYS 12 V dal BUS_POWER (pin 24–26) -> eFuse TPS259470 (3 A, UVLO/OVLO, blocco inverso) -> buck TPS56637 12 -> 5 V / 6 A -> 5V_EXT

4) Power mux TPS2121: IN1 5V_EXT (jack o VIN_SYS) con priorita, IN2 VBUS USB-C. Uscita 5V_SYS verso AXP313A e 3V3_PERI

TPS56637 (tabella 4 del datasheet, Vout 5 V): Vfb 0.6 V, R103/R104 73.2k/10k = 4.99 V, L 3.3 uH, Co 4x22 uF, fsw 600 kHz, diodo forward-forward R116 20k + C126 100 pF.

PR1: IN1 sotto 4.49 V -> passa a IN2. OV1/OV2: canale escluso sopra 5.98 V. CP2 a GND (modo VREF). ILM 22.1k -> 4–5 A. SS 100 nF.

EN: 169k/36.1k (schema tipico TI). MODE libero = FCCM, R114 DNP per Eco-mode. Niente ideal diode: il blocco inverso lo fa l'eFuse, il jack entra con lo Schottky D103.

eFuse: RILM 1.11k = 3 A (333/RILM). OV1/OV2 (2.4 V/615k/62k) = 9.5–17 V. Tempi 22 nF. Il jack deve erogare corrente per sostenere il picco di boot (spunto 1–2 s).

5) SY6280AAC: VBUS in uscita verso una periferica (source, -1 A). Ingresso da 5V_EXT: si accende solo con jack o VIN_SYS presenti

3) USB-C (HRO TYPE-C-31-M-12): solo USB 2.0, FUSB302B per CC/PD, TVS SMF5.0A su VBUS (nessun fusibile), ESD TPD2EUSB30A su CC1/CC2

Ilim = 6800 / R112 = 1 A. R113: uscita spenta finche Linux non abilita VBUS_SRC_EN (comandato da tcpm).

FUSB302B: I2C condivisa con AXP313A ed espansore su S_TWI0 (PL0/PL1, 1.8 V: VIH 1.32 V, pull-up 1.71 V..VDD da datasheet). Pull-up 2.2k e INT_N 10k a VCC-1V8 sul foglio CPU.

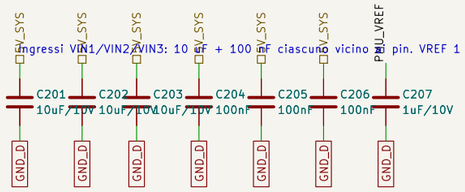
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/10

PMIC AXP313A (datasheet X-Powers v1.0) come su Orange Pi Zero3 (H618 + AXP313A): VIN 3.0–5.5 V da 5V_SYS, I2C 0x36 su S_TWI0 (PL0/PL1). Serie 2xx.

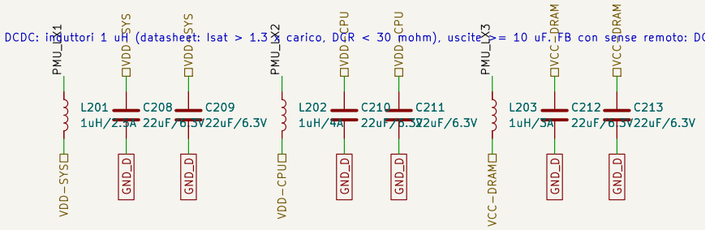
DCDC1 VDD-SYS 0.9 V (VDD_SYS + VDD_GPU, max 0.99 V) 1.5 A, DCDC2 VDD-CPU 0.81–1.1 V 3 A, DCDC3 VCC-DRAM 1.1 V (LPDDR4 VDD2/VDDQ) 2 A,

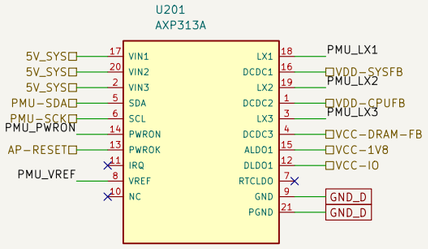
ALDO1 VCC-1V8 300 mA (VCC_PLL, VDD18_DRAM, DCXO, AVCC, VCC_PC/eMMC VCCQ, LPDDR4 VDD1), DLDO1 VCC-IO 3.3 V 500 mA (VCC_IO/PG/PI/USB). RTCLDO non usato (H616 senza VCC_RTC).

Ingressi VIN1/VIN2/VIN3: 10 uF + 100 nF ciascuno vicino ai pin. VREF 1 uF (datasheet).

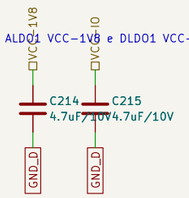


DCDC: induttori 1 uH (datasheet: Isat > 1.3x carico, DCR < 30 mohm), uscite V = 10 uF + FB con sense remoto: DCDC1 da VDD_GPUFB (F8), DCDC2 da VDD_CPUFB (L8), DCDC3 dalla LPDDR4.

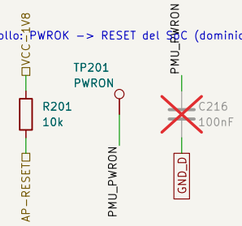




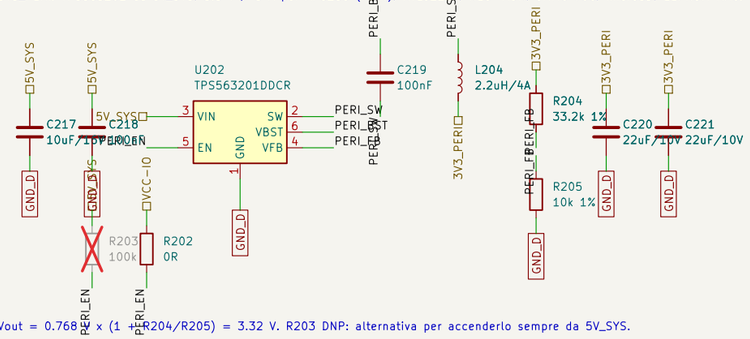
LDO: ALDO1 VCC-1V8 e DLDO1 VCC-IO con 4.7 uF (datasheet: >= 1 uF). RTCLDO libero (LDO non usato: pin flottante, nessun condensatore).



Controllo: PWROK -> RESET del SoC (dominio VCC_PLL 1.8 V), pull-up 10k a VCC-1V8. Accensione all'arrivo di VIN; PWRON su test point. IRQ non collegato (nessun GPIO libero).



3V3_PERI: TPS563201 da 5V_SYS, 3.3 V / 3 A per AP6256 (VBAT), RTL8211F, NEO-M9N, eMMC VCC, microSD, EEPROM MAC. EN da VCC-IO (parte dopo DLDO1).



Vout = 0.768 V x (1 + R204/R205) = 3.32 V. R203 DNP: alternativa per accenderlo sempre da 5V_SYS.

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MEMORIE: LPDDR4 2 GB x32 (1 chip, come Orange Pi Zero3), eMMC 8 bit su SMHC2 (PC, 1.8 V), microSD su SMHC0 (PF, 3.3 V). Serie 4xx.

U401 MT53E512M32D1ZW-046 IT:B: 1 die, 2 canali x16, CA comuni ai due canali. VDD1 = VCC-1V8 (ALDO1), VDD2/VDDQ = VCC-DRAM 1.1 V (DCDC3). Niente VTT/VREF/VPP.

CA0 <- SA1, CA1 <- SCS0, CA2 <- SA0, CA3 <- SCS1, CA4 <- SA4, CA5 <- SA3, CS0 <- SA2, CS1 <- SA5 (da phy_init U-Boot, NON VERIFICATO). DQ dritti: canale A = byte 0-1, canale B = byte 2-3.

ODT_CA_A/B a VDD2 con 100R; ZQ0/ZQ1 240R 1% a VDDQ (come il reference Allwinner). CS1/CKE1 su sfere NC del chip a singolo rank.

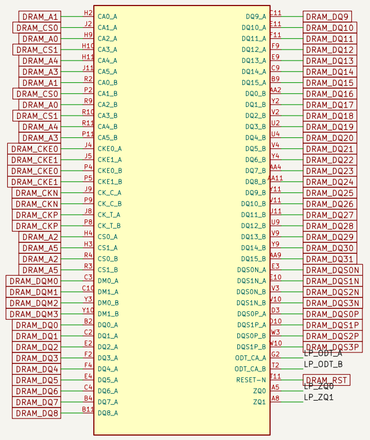
Disaccoppiamento LPDDR4: VDD2/VDDQ 4x1uF + 6x0.1uF, VDD1 2x1uF + 2x0.1uF (da rifinire con la PDN Micron).

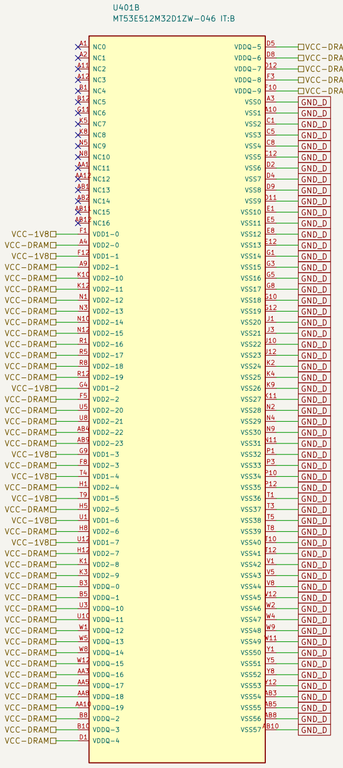
R405 sense del DCDC3 dell'AXP313A preso sulle sfere VDD2 della LPDDR4.

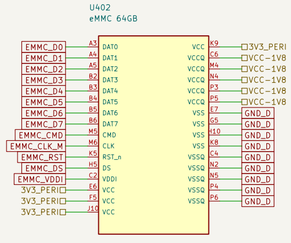
U402 eMMC 64 GB FBGA-153 (modello da scegliere): VCC = 3V3_PERI, VCCQ = VCC-1V8 (VCC_PC 1.8 V: HS200, DS per HS400). CLK 33R serie, pull-up 10k su CMD/DAT0-7/RST_n a VCCQ.

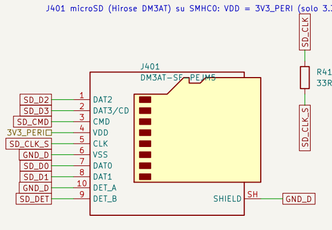
J401 microSD (Hirose DM3AT) su SMHC0: VDD = 3V3_PERI (solo 3.3 V, niente UHS). CLK 33R serie, pull-up 47k a 3V3_PERI. SD_DET sull'espansore (P15), 10k a VCC-IO.

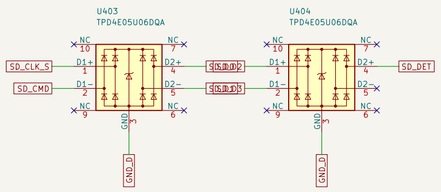
ESD TPD4E05U06DQA vicino allo slot: CLK (lato slot), CMD, DAT0-3, DET.











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RETE: PHY Realtek RTL8211F-CG (datasheet rev. 1.7) su RGMII EMAC0 banco PI (VCC_PI 3.3 V). Serie 5xx. Net RGMII_*/PHY_* globali verso il foglio CPU.

Strap (4.7k): PHYAD = 001, PLLOFF = 0, TXDLY = 1 e RXDLY = 1 (rgmii-id, ritardi nel PHY), CFG_EXT = 1 con CFG_LDO = 00: pad RGMII alimentati da fuori a 3.3 V.

U501
RTL8211F-CG

PHY_TXC_P	20	TXC	MDIP0 1 MDIO0_P
PHY_TXD0_P	18	TXD0	MDIN0 2 MDIO0_N
PHY_TXD1_P	17	TXD1	MDIP1 4 MDI1_P
PHY_TXD2_P	16	TXD2	MDIN1 5 MDI1_N
PHY_TXD3_P	15	TXD3	MDIP2 6 MDI2_P
PHY_TXCTL_P	19	TXCTL	MDIN2 7 MDI2_N
RGMII_RXCK	27	RXC/PHYAD1	MDIP3 9 MDI3_P
RGMII_RXD0	25	RXD0/RXDLY	MDIN3 10 MDI3_N
RGMII_RXD1	24	RXD1/TXDLY	LED0/CFG_EXT 32 PHY_LED0
RGMII_RXD2	23	RXD2/PLLOFF	LED1/CFG_LDO0 33 PHY_LED1
RGMII_RXD3	22	RXD3/PHYAD0	LED2/CFG_LDO1 34 PHY_LED2
RGMII_RXCTL	26	RXCTL/PHYAD2	RSET 39 PHY_RSET
PHY_MDC	13	MDC	REG_OUT 30 PHY_REG_OUT
PHY_MDIO	14	MDIO	DVDD10 21 PHY_VDD10
PHY_INT_N	31	INTB/PMEB	AVDD10 3 PHY_VDD10
PHY_RST_N	12	PHYRSTB	AVDD10 8 PHY_VDD10
PHY_CLKOUT	35	CLKOUT	AVDD10 38 PHY_VDD10
PHY_XI	36	XTAL_IN	AVDD33 11 3V3_PERI
PHY_XO	37	XTAL_OUT	AVDD33 40 3V3_PERI
			DVDD33 29 3V3_PERI
			DVDD_RG 28 3V3_PERI
			EPAD_GND 41 GND_D

22R serie verso il SoC su TXC, TXD0-3, TXCTL (MAC → PHY). RX senza serie (sorgente PHY).

RGMII_TXCK	RGMII_TXD0	RGMII_TXD1	RGMII_TXD2	RGMII_TXD3	RGMII_TXCTL	
R501 22R	R502 22R	R503 22R	R504 22R	R505 22R	R506 22R	
PHY_TXC_P	PHY_TXD0_P	PHY_TXD1_P	PHY_TXD2_P	PHY_TXD3_P	PHY_TXCTL_P	

Strap su RXD3/RXC/RXCTL e LED (4.7k). LED1/LED2 attivi alti verso i LED del RJ45 (330R); LED0 solo strap CFG_EXT.

3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI	PHY_LED1	PHY_LED2	
R507 4.7k	R508 4.7k	R509 4.7k	R510 4.7k	R511 4.7k	R512 4.7k	R513 4.7k	R514 4.7k	R515 4.7k	
RGMII_RXD3	GND_D	GND_D	GND_D	RGMII_RXD1	RGMII_RXD0	PHY_LED0	GND_D	GND_D	
RGMII_RXCK	RGMII_RXCTL	RGMII_RXD2							

MDIO 1.5k a 3.3 V, INT 4.7k, PHYRSTB 10k + 100 nF (il PHY esce dal reset da solo, Linux puo' ripeterlo dall'espansore). RSET 2.49k 1%.

3V3_PERI	3V3_PERI	3V3_PERI	PHY_RST_N	PHY_RSET	PHY_CLKOUT	
R516 1.5k	R517 4.7k	R518 10k	C501 100nF	R519 2.49k 1%	R520 0R	
PHY_MDIO	PHY_INT_N	PHY_RST_N	GND_D	GND_D	RGMII_CLKIN	

R520 DNP: CLKOUT 125 MHz verso RGMII_CLKIN (PI13), non stato di default. PHY_RST_N e PHY_INT_N passano dall'espansore TCA6416A (foglio CPU).

Quarzo 25 MHz +-50 ppm (requisito datasheet) con 2x 15 pF. Switching regulator 1.0 V: REG_OUT → 2.2 uH → AVDD10/DVDD10.

PHY_REG_OUT	PHY_VDD10	PHY_VDD10	PHY_VDD10	PHY_VDD10	PHY_VDD10	3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI	3V3_PERI		
L501 2.2uH	C504 22uF	C505 0.1uF	C506 0.1uF	C507 0.1uF	C508 0.1uF	C509 10uF	C510 0.1uF	C511 0.1uF	C512 0.1uF	C513 0.1uF		
PHY_VDD10	GND_D	GND_D	GND_D	GND_D	GND_D	GND_D	GND_D	GND_D	GND_D	GND_D		

PHY_XI	Y501 25MHz	PHY_XO		
C502 15pF	GND_D	C503 15pF		
GND_D		GND_D		

RJ45 HanRun HR911130C (100/1000Base-T, magnetici e LED integrati). Prese centrali lato PHY (P1, comuni) a GND con 100 nF. P10 = rete Bob Smith verso lo schermo.

J501
HR911130C

MDI0_P	P2	MDI0+	LED_G_A 11 ETH_LED1_A
MDI0_N	P3	MDI0-	LED_G_K 12 GND_D
MDI1_P	P4	MDI1+	LED_Y_A 14 ETH_LED2_A
MDI1_N	P7	MDI1-	LED_Y_K 13 GND_D
MDI2_P	P5	MDI2+	BOB_SMITH P10 GND_D
MDI2_N	P6	MDI2-	SHIELD0 GND_D
MDI3_P	P8	MDI3+	SHIELD1 GND_D
MDI3_N	P9	MDI3-	
ETH_CT	P1	CT	

ETH_CT → C514 100nF → GND_D

LED verde = LED1 del PHY (100M), giallo = LED2 (1000M), entrambi attivi alti.

PHY_LED1 → R522 510R → ETH_LED1_A
PHY_LED2 → R523 510R → ETH_LED2_A

EEPROM MAC 25AA02E48 (EUI-48 per la MAC dell'EMAC0) su SPI1 locale (prima dei 33R verso il bus), CS dall'espansore P16 con 10k a 3V3_PERI (deselezionata al reset).

Pinout SOT-23-6 dal datasheet Microchip DS20002123G, tabella 4-1: 1 SCK, 2 VSS, 3 SI, 4 SO, 5 CS, 6 VCC.

U502
25AA02E48T-I/OT

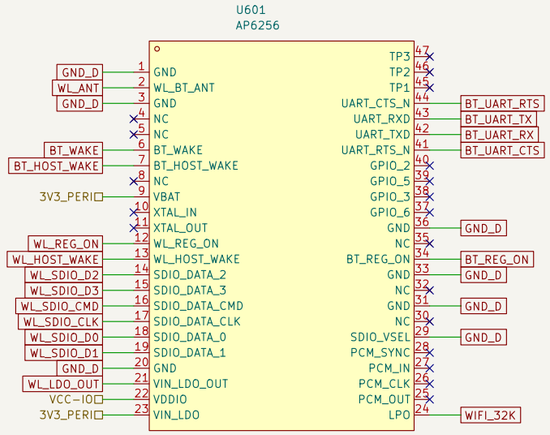
3V3_PERI	6	VCC	SO 4 SPI1_MISO
PHY_EE_CS_N	5	CS	VSS 2 GND_D
SPI1_CLK	1	SCK	
SPI1_MOSI	3	SI	

3V3_PERI → R521 10k → PHY_EE_CS_N
3V3_PERI → C515 0.1uF → GND_D

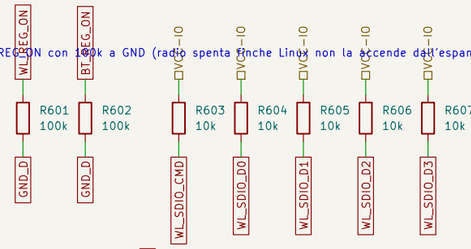
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File: rete.kicad_sch		
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/10

RADIO: AMPAK AP6256 (Wi-Fi 5 + BT 5.0) su SMHC1 PG0–PG5 (SDIO), UART1 PG6–PG9 con RTS/CTS, REG_ON e wake sull'espansore TCA6416A. Serie 6xx. Net WL_*/BT_*/WIFI_32K globali.

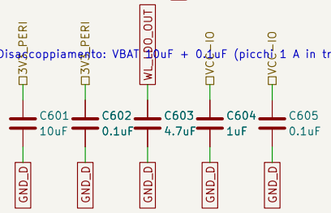
VDDIO = VCC-IO 3.3 V (= VCC_PG), SDIO 2.0 fino a 50 MHz. VBAT e VIN_LDO da 3V3_PERI. LPO 32.768 kHz da X32KFOUT (PG10), derivato dal 24 MHz: da verificare.



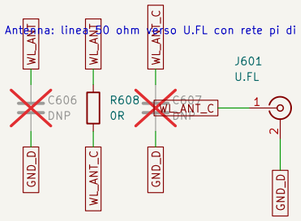
REG_ON con 100k a GND (radio spenta finché Linux non la accende dall'espansore). Pull-up 10k su CMD e DATO-3 a VCC-IO.



Disaccoppiamento: VBAT 10uF + 0.1uF (picchi 1 A in trasmissione), VIN_LDO_OUT 4.7uF, VDDIO 1uF + 0.1uF.



Antenna: linea 50 ohm verso U.FL con rete pi di adattamento (OR + 2x DNP) da tarare.



UART incrociata: TXD del modulo -> BT_UART_RX (PG7), RXD <- BT_UART_TX (PG6), RTS_N -> BT_UART_CTS (PG9), CTS_N <- BT_UART_RTS (PG8).

PCM (25–28) non collegati: niente audio Bluetooth. XTAL_IN/OUT liberi (quarzo interno). GPIO e TP liberi.

Sheet: /Radio/
File: radio.kicad_sch

Title:

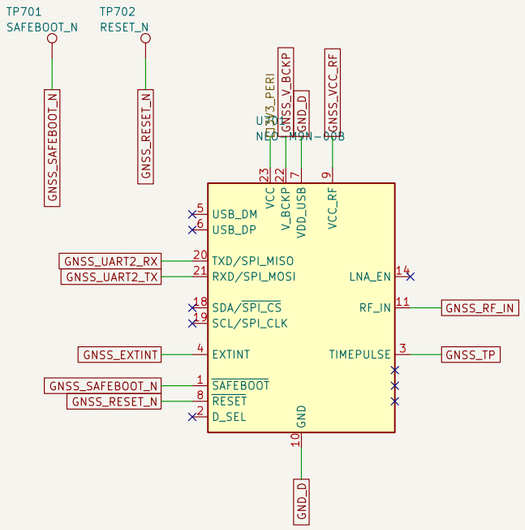
Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 8/10

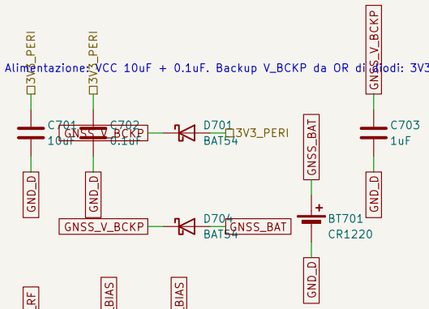
GNSS: u-blox NEO-M9N-00B (formato NEO, stesso footprint del NEO-F10T). Pin dal datasheet UBX-19014285 R08, tabella 9 (= simbolo NEO-M9N di KiCad). Serie 7xx.

UART2: TXD del modulo -> GNSS_UART2_RX (PG16), RXD <- GNSS_UART2_TX (PG15). TIMEPULSE -> 33R -> GNSS_PPS (PG13, EINT). EXTINT (pin 4, PIO7, time mark TIM-TM2) letto anche da PG14 (EINT, solo ingresso).

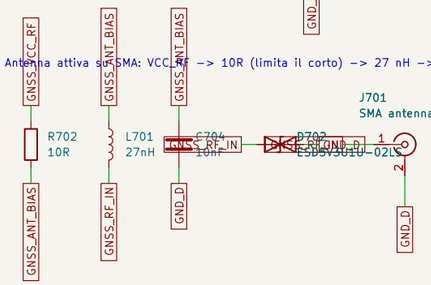
D_SEL (pin 2) aperto = UART + I2C. V_USB (pin 7) a GND: USB non usata (da confermare sull'integration manual). LNA_EN e riservati liberi.



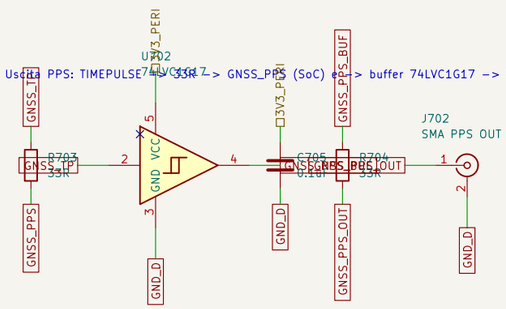
Alimentazione: VCC 10uF + 0.1uF. Backup V_BCKP da OR di diodi: 3V3_PERI (BAT54) e pila CR1220 (BAT54), nessuna ricarica della pila.



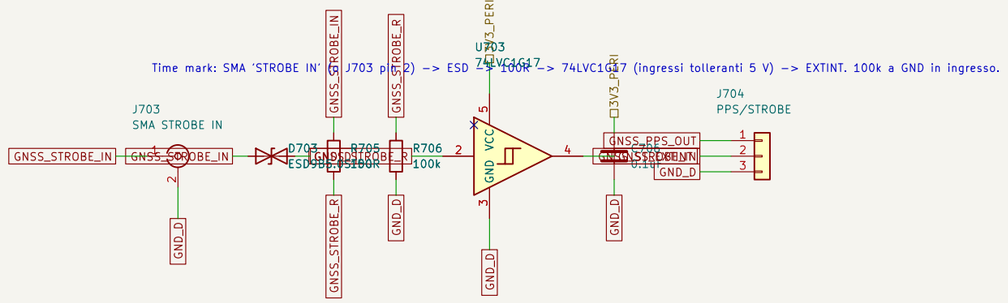
Antenna attiva SMA: VCC_RF -> 10R (limita il corto) -> 27 nH -> RF_IN. ESD a bassa capacita sul centrale. Linea 50 ohm corta.



Uscita PPS: TIMEPULSE -> 33R -> GNSS_PPS (SoC) e -> buffer 74LVC1G17 -> 33R -> SMA 'PPS OUT' e header J703. Nessun pull sul TIMEPULSE.



Time mark: SMA 'STROBE IN' -> J703 p(1,2) -> ESD -> 100R -> 74LVC1G17 (ingressi tolleranti 5 V) -> EXTINT. 100k a GND in ingresso.



PWR_FLAG

GNSS_V_BCKP

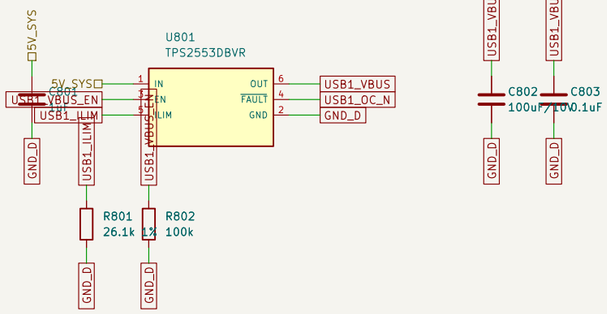
Sheet: /GNSS/		
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/10

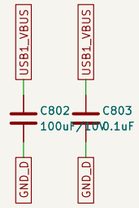
USB HOST: USB1 del SoC su presa USB-A (C46407), solo host. Serie 8xx. Net USB1_* globali verso il foglio CPU.

La porta non alimenta mai la scheda: VBUS esce solo dal load switch TPS2553.

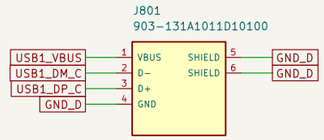
TPS2553: EN attivo alto dall'espansore (P03) con 100k a GND, FAULT open-drain su PC12 (pull-up 10k a VCC-1V8 sul foglio CPU).

RILIM 26.1k 1% = 0.91 / 0.99 / 1.08 A (min/nom/max, tabella 2 del datasheet).

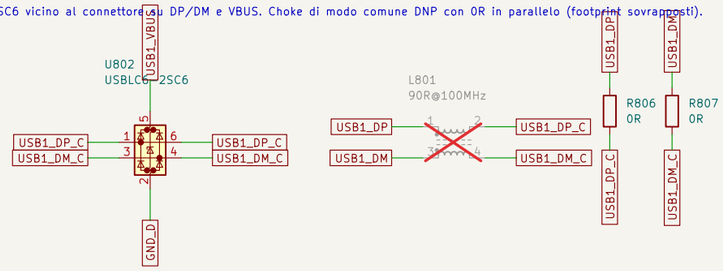




J801 USB-A (903-131A1011D10100 / C46407), schermo a GND.



ESD USBLC6-2SC6 vicino al connettore su DP/DM e VBUS. Choke di modo comune DNP con 0R in parallelo (footprint sovrapposti).



Sheet: /USB host/ File: usb.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 10/10